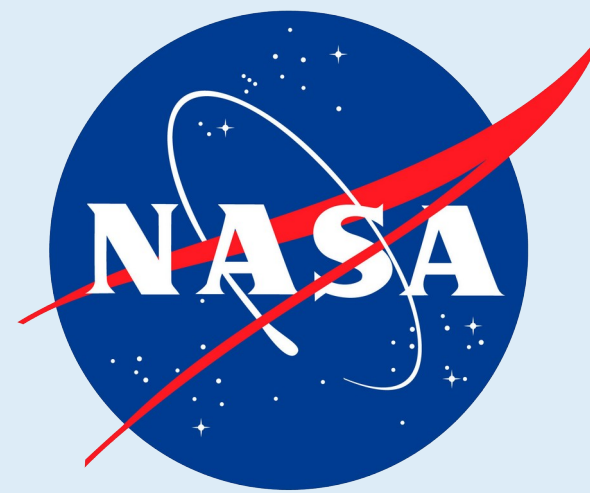




Cross-mission expiMap analysis recovers established tissue responses and identifies complementary pathway shifts in mouse spaceflight transcriptomes

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ABSTRACT

Mission differences can obscure spaceflight responses shared across studies. We mapped NASA OSDR mouse bulk RNA sequencing samples to tissue-matched ARCHS4 references. Reactome-constrained expiMap models analyzed about 2,000 highly variable genes. Flight-ground shifts were balanced across projects and evaluated with gene-set enrichment, project holdouts, three complete training runs, composition proxies, and member-gene review. Literature review followed scoring. Retained programs were lower for thymic repair, skin maintenance, liver adaptive immunity, and splenic adaptive and innate immunity. Spleen showed the strongest multi-pathway result.

INTRODUCTION

PROJECT OBJECTIVE

Identify gene programs that shift consistently across spaceflight missions.

THE CENTRAL CHALLENGE

Observed expression **Spaceflight biology** + **Tissue biology** + Animal variation + Study / protocol

Study identity and protocol can dominate an unconstrained expression representation. Tissue-matched reference mapping, accession conditioning, and equal project weighting reduce this bias without claiming to remove all mission confounding.

FINAL ANALYSIS SCOPE

Tissue	ARCHS4	OSDR	Projects	Programs
Thymus	1,362	117	5	387
Skin	2,593	151	4	319
Liver	5,000	197	9	364
Spleen	6,289	100	5	360

OSDR counts are primary analysis samples. Spleen excludes one condition-strain-confounded project.

ROBUSTNESS GATE

ssGSEA + GSEA

Held-out projects

3 full trainings

Composition proxies

Member genes

Literature review

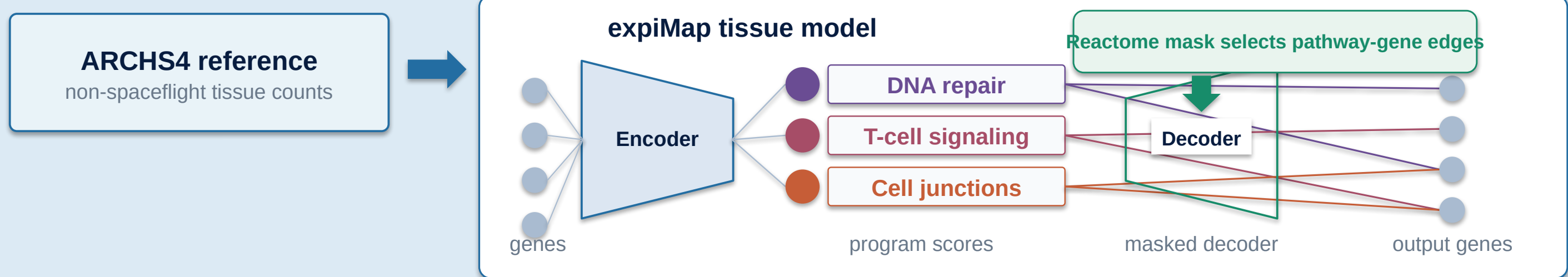
Complementary programs add a plausible perspective beyond the dominant phenotype in prior literature. De novo nodes were not retained in the final models.

METHODS

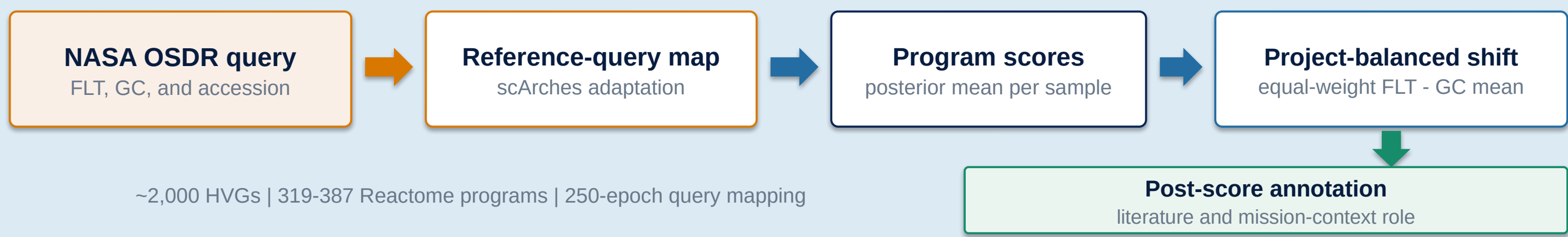
Pathway structure is wired into the decoder

Train a tissue reference, then map flight and ground samples into the same annotated program space.

1 REFERENCE TRAINING

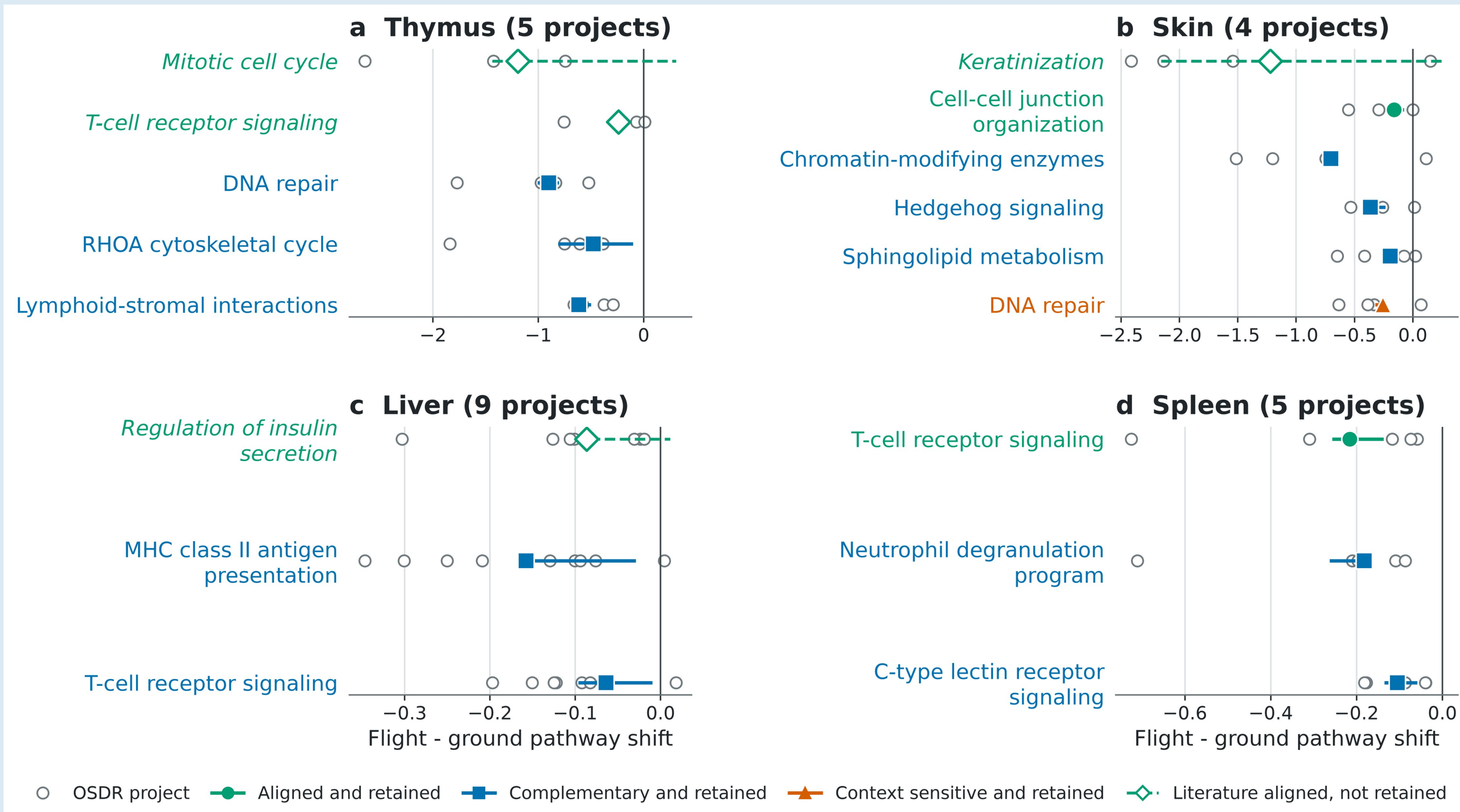


2 QUERY MAPPING AND CROSS-MISSION COMPARISON



~2,000 HVGs | 319-387 Reactome programs | 250-epoch query mapping

RESULTS: ALIGNED ANCHORS AND RETAINED PROGRAM SHIFTS

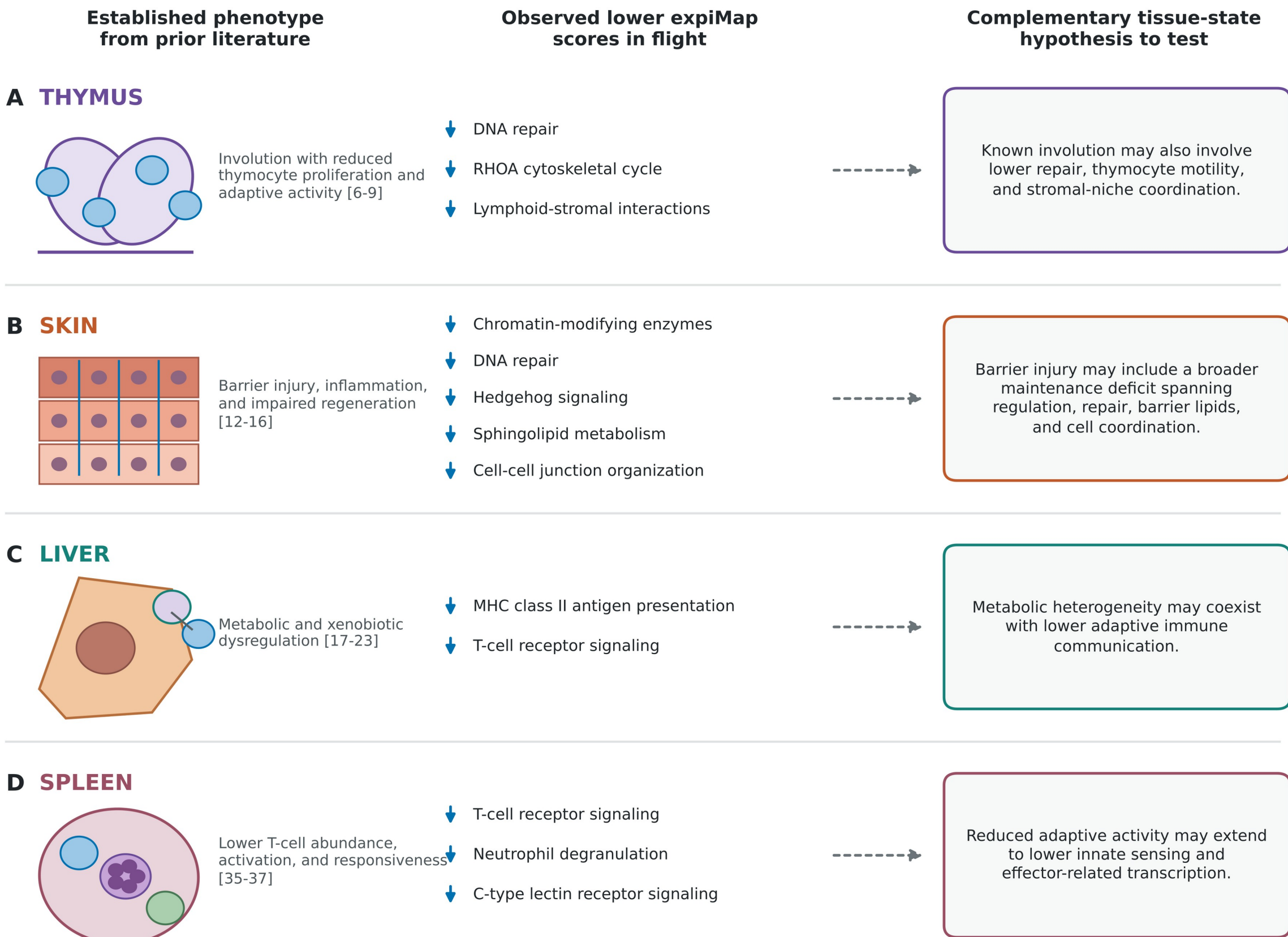


Filled markers show 13 retained programs. Open diamonds are literature-aligned context only, not retained findings; keratinization reversed in one of three trainings. Colored ranges span three complete trainings; axes are tissue specific.

RESULTS: BIOLOGICAL INTERPRETATION

Prior evidence, observed shifts, and hypotheses to test

Prior literature, observed pathway shifts, and hypotheses to test



Prior literature, this study's observations, and new hypotheses are shown separately; dotted arrows denote inference, not causality. Lower scores do not prove pathway inhibition, and cell composition may contribute.

CONCLUSIONS

- Spleen was strongest: T cell receptor, neutrophil degranulation, and C-type lectin receptor programs were lower across five projects and three trainings; all had GSEA FDR <0.05.
- Skin and thymus emphasized lower tissue maintenance programs; liver showed lower adaptive immune signaling amid heterogeneous metabolism.
- Bulk latent scores are testable hypotheses, not causal or cell-resolved measures of pathway activity.

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github.com/jasont314/nasa-mouse